Corrections and checks (red). Wilson-line index order: $U^{\dagger\,bc} = U^{cb}$. The unmeasured gluon at z created after the shock carries $U^{cb}(z)$, and a rotated source is $U^{ce}(x)\rho^e(x)$; five remaining $U^{e'c'}(z)$ in Group III (Rows III, V, VI) are now $U^{c'e'}(z)$. Every row was checked against its source in Groups I–III: (source) $= \log\frac{\vee}{k^+}\times$(BFKL) $+\log\frac{k^+}{\Lambda}\times$(JIMWLK) + (this file), using $\log\frac{(\vee-k^+)X+k^+W}{\Lambda X+k^+W} \to \log\frac{\vee}{k^+} + \log\frac{(1-k^+/\vee)X+(k^+/\vee)W}{W}$ and the plus-prescription endpoints. All rows agree except Group I, Row III, N_{3b}, where one term was missing (added below). Every term is finite as $\Lambda \to 0$ and has a finite limit as $\vee \to \infty$. Group II, Row IV needs the definition of $\mathcal{P}_{gg}$ given there.

Finite terms $(\Lambda \to 0)$

1

2 Group I (with c.c.)

2.1 Row I

NA

2.2 Row II

We write it in two parts:

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\int_{w,w',x,x',} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma-\log\left(\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]\times\log\left(1-\frac{k^+}{\vee}\right)\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^{b}(x)-U^{ab}(x)\,\rho^{b}(x)\Big]\\ &+\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\int_{w,w',x,x',} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left\{-\frac{67}{9}+\frac{\pi^2}{3}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\,\log\left[\frac{\vee}{k^+}\right]\right\}\times\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^{b}(x)-U^{ab}(x)\,\rho^{b}(x)\Big]\end{aligned}\tag{2.1}$$

$$\begin{aligned}\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y} e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\;R^i(\mathbf{k},\mathbf{p})\\ &\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{hbc}\,U^{ah}(w)\,\big\{\rho^b(x),\rho^c(y)\big\}-f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\big\{\rho^d(x),\rho^e(y)\big\}\Big]\end{aligned}\tag{2.2}$$

where

$$R^i=-2k^i\log\left(\frac{p^2}{k^2+\frac{k^+}{\vee}\,p^2}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\,\log\left[\frac{k^2}{(k-p)^2}+\frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right)\right]-\frac{p^2k^i+k^2p^i}{(k-p)^2}\,\log\left(1-\frac{k^+}{\vee}\right).\tag{2.3}$$

2.3 Row III

We write it in three parts:

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left\{\right.\\ &-\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-z)^2+(x-w)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-w)^2+(x-z)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}\times\log\left|\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right|\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-w)^2+(x-z)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}\times\log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right|\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\\ &-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left|\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right|\\ &\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\left.\right\}\end{aligned}\tag{2.4}$$

$$
\begin{aligned}
\frac{d^3 N_{2\,\mathrm{NLO}}}{d^3 k} = &- \frac{1}{(2\pi)^3} \int_{w,w'} e^{-ik(w'-w)} \Bigg\{ \\
&\frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+}\, \log\left(1 - \frac{k^+}{\vee}\right) \int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+}\, \log\left(1 - \frac{k^+}{\vee}\right) \int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left|\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right| \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left|\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right| \frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left|\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right| \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left|\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right| \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]\,(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+ \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left|\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right| \frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&- \frac{ig^4\, f^{abd}}{8\pi^5}\, \frac{1}{k^+} \int_{x,y,z,x'} \log\left|\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right| \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \Bigg\}
\end{aligned}
\tag{2.5}
$$

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right](x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left|\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right| \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,\left[(x-w)^2-(x-z)^2\right]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\Bigg\}
\end{aligned}
\tag{2.6}
$$

$$
\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \mathcal{A}_0\,\frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right] \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& +\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right] \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\int_{x,y,z,x'} \log\left[\frac{(z-w)^2}{(x-w)^2}\right] \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\right]\Bigg\}
\end{aligned}
\tag{2.7}
$$

where:

$$\mathcal{A}_0 = \frac{1}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \left\{ \arctan\left[\frac{(x-z)^2 - (x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \right] + \arctan\left[\frac{(x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \right] \right\} \tag{2.8}$$

Simplification (added): closed form of $\mathcal{A}_0$. With $p = x-w$, $q = x-z$ one has $\sqrt{(x-w)^2(x-z)^2 - [(x-w)\cdot(x-z)]^2} = |p \wedge q|$ and, by the arctan addition formula,

$$\mathcal{A}_0 \;=\; \frac{\theta}{\left|(x-w)\wedge(x-z)\right|}\,, \qquad \theta = \angle\big(x-w,\; z-w\big) \in (0,\pi)\,, \tag{2.9}$$

i.e. $\mathcal{A}_0$ is the angle subtended at w by the segment $\overline{xz}$, divided by twice the area of the triangle (x,w,z) (exact identity, verified numerically). In this form N_{3a}, N_{3b} are manifestly finite and rotation invariant, and the collinear limits (w on the line xz) are transparent.

Added term (red). The last term of N_{3b} was missing. In Group I, Row III, N_{3b} has four terms; the fourth one, with kernel $L_4\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$ and prefactor $-\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}$, was dropped. Here $L_4 = \log\frac{[k^+(x-w)-(k^+-\Lambda)(x-z)]^2}{[k^+(x-w)-\Lambda(x-z)]^2} \to \log\frac{(z-w)^2}{(x-w)^2}$ as $\Lambda \to 0$. With it, Row III satisfies (source) $= \log\frac{\vee}{k^+}\times$(BFKL) $+\log\frac{k^+}{\Lambda}\times$(JIMWLK) + (finite) exactly (checked numerically at random points); without it the finite part is off by an $O(1)$ amount. The logarithms and $\mathcal{A}_0$ in N_{3a}, N_{3b} depend on x,z,w, so they were moved inside $\int_{x,y,z,x'}$ (red).

2.4 Row IV

$$\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)} \Bigg\{ & \\
& \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right] \times \left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\
& \left. - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2}\right] \\
& + \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big]
\end{aligned} \tag{2.10}$$

2.5 Row V

First part

$$\begin{aligned}
\frac{d^3N_{1\,\rm NLO}}{d^3k} = & \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \\
& \left[\delta_{im}\,\delta_{jk} - \left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times \\
& \Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big] \\
& - \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\left[\frac{1}{q^+}\right]_+ \\
& \times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\Big]
\end{aligned} \tag{2.11}$$

Second part

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\int_{k^+}^0\frac{dq^+}{2\pi}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{\left[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left\{\left[\frac{k^+}{q^+}\right]_+\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{2.12}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+(k^+-p^+)}{k^{+2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\,\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\right]\\
&\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\,\left[\frac{k^+}{q^+}\right]_+\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{11}{6}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{2.13}
$$

3 Group II

3.1 Row I

3.1.1 First part

$$\frac{d^3N_{1a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times$$
$$\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}$$
$$\left[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{c'b}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{3.1}$$

$$\frac{d^3N_{1b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{$$
$$\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w')^2}{(x-w')^2}\right]\times$$
$$\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]$$
$$+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}$$
$$\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{c'b}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{cb}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right] \tag{3.2}$$

3.1.2 Second part

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\Bigg\{$$
$$\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times$$
$$\Bigg[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}$$
$$+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}$$
$$+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}$$
$$-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\Bigg]\times$$
$$\left[\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]-\log\left[\frac{(x-w)^2}{(x'-w')^2}\right]\right]\Bigg\}\times$$
$$\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)$$
$$-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\Bigg] \tag{3.3}$$

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$

$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$

$$\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times$$

$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$

$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]$$

$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$

$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$

$$\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times$$

$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$

$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]$$

$$+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\;-(x-w)^2\,(x'-z)^2\,(x'-w')^2]}\times$$

$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$

$$\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]$$

$$\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.$$

$$\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \tag{3.4}$$

$$
\begin{aligned}
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{3.5}
$$

$$
\begin{aligned}
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} &= +\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{\left[(x-z)^2-(x-w)^2\right]}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{\left[(x'-w')^2-(x'-z)^2\right]}\,\frac{1}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\\
&\left[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{3.6}
$$

$$
\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\left[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\right.\\
&\left.+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\right]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.\\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right]
\end{aligned}
\tag{3.7}
$$

$$
\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\frac{g^4\,d_\perp}{32\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]}\\
&\frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&+\frac{1}{(2\pi)^3}\frac{g^4\,d_\perp}{32\pi^5}\left[\frac{1}{\vee}-\frac{1}{k^+}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{(z-w)^2\,(z-w')^2}\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&+\frac{1}{(2\pi)^3}\frac{g^4\,d_\perp}{32\pi^5}\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}\\
&\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\,\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{cb}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{c'b}(z)\,U^{d'e'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{de'}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{3.8}
$$

Note (N_{2f}). Each of the three terms of N_{2f} behaves like $1/(z-w)^2$ as $z\to w$ (and like $1/(z-w')^2$ as $z\to w'$), which is not integrable. Their sum is regular: the $1/(z-w)^2$ coefficients $-(1-k^+/\vee)/k^+$ (first term) and $\left(\frac{1}{\vee}-\frac{1}{k^+}\right)$ (second term) cancel exactly (*checked: the cancellation is algebraically exact, the two coefficients are equal and opposite identically, not merely numerically*). The three terms must therefore be integrated together, not separately.

3.2 Row II

NA

3.3 Row III

NA

3.4 Row IV

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = &\underbrace{-\frac{g^2}{8\pi^2}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\left[\mathcal{P}_{gg}(\xi)\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}+2\,N_c\,\frac{(1-\xi)}{\xi}\,\frac{d^3N^{\mathrm{sub}}_{\xi}}{d^3k}\right]}_{\mathrm{DGLAP}}\\
&\underbrace{+\frac{g^2\,\beta_0}{16\pi^2}\left(\frac{2}{\epsilon}+2\,\gamma_E+\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right)\frac{d^3N_{\mathrm{LO}}}{d^3k}}_{\mathrm{RC}}\\
&-\frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\
&\left[U^{ce'}(z)-U^{ce'}(x')\right]\times\left[U^{ce}(z)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)\\
&+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-2\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\right]\times\\
&\left[U^{ce'}(y')-U^{ce'}(x')\right]\times\left[U^{ce}(y)-U^{ce}(x)\right]\rho^{e'}(x')\rho^{e}(x)
\end{aligned}
\tag{3.9}
$$

where:

$$\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\;\rho^{e'}(x')\rho^{e}(x) \tag{3.10}$$

Corrected subtraction term (replaces $d^3N^y_{\xi\,\mathrm{LO}}$). The $2N_c\frac{1-\xi}{\xi}$ part of $\mathcal{P}_{gg}$ produces a hidden $\log(\vee/k^+)$ from $\xi\to 0$, where $y,y'\to z$ at fixed w,w', since $y-z=\xi(w-z)$. Its $\xi\to 0$ limit must be subtracted *exactly*:

$$\begin{aligned}\frac{d^3N^{\mathrm{sub}}_{\xi}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\pi}\,\frac{(y-z)\cdot(y'-z)}{(y-z)^2\,(y'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\ \times\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,\big[U^{ce}(z)-U^{ce}(x)\big]\,\rho^{e'}(x')\,\rho^{e}(x).\end{aligned} \tag{3.11}$$

It is IR and UV finite in $d=2$, contains no μ, and is symmetric under $(x,y)\leftrightarrow(x',y')$. Changing variables $y=z+\xi(w-z)$, $y'=z+\xi(w'-z)$ (Jacobian ξ^4, $K(y,y',z)=K(w,w',z)/\xi^2$) gives exactly

$$\begin{aligned}\frac{g^2}{8\pi^2}\int_{k^+/\vee}^1\frac{d\xi}{\xi^2}\,2N_c\,\frac{1-\xi}{\xi}\,\frac{d^3N^{\mathrm{sub}}_{\xi}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^5}\,\frac{1}{k^+}\Big[\log\frac{\vee}{k^+}-\Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}K(w,w',z)\,K(x,x',z)\\ \times\Big[U^{ce'}(z)-U^{ce'}(x')\Big]\,\big[U^{ce}(z)-U^{ce}(x)\big]\,\rho^{e'}(x')\,\rho^{e}(x),\end{aligned} \tag{3.12}$$

that is, the BFKL piece (BFKL file, G-II R-IV) plus the $(1-k^+/\vee)$ term above. The $(1-\xi)/\xi$ part of the DGLAP bracket is then manifestly symmetric and μ-free apart from the collinear pole:

$$\begin{aligned}2N_c\frac{1-\xi}{\xi}\Big[L_\epsilon\,\frac{d^3N_{\xi\,\mathrm{LO}}}{d^3k}+\frac{d^3N^{\mathrm{sub}}_{\xi}}{d^3k}\Big] = -\frac{2N_c(1-\xi)}{\xi}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\\ \times\int_{x,y,x',y',z} e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\,\frac{1}{\pi}\,K(y,y',z)\Big[\mathcal{K}(x-y)\cdot\mathcal{K}(x'-y')\,W(y,y')-K(x,x',z)\,W(z,z)\Big],\end{aligned} \tag{3.13}$$

with $L_\epsilon=\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}$, $W(u,u')=[U^{ce'}(u')-U^{ce'}(x')][U^{ce}(u)-U^{ce}(x)]\rho^{e'}(x')\rho^{e}(x)$, and the first z-integral taken in $d=2-\epsilon$ (it equals $-\pi L_\epsilon$ times $\mathcal{K}\cdot\mathcal{K}\,W$). At fixed w,w' the square bracket is $O(\xi)$ as $\xi\to 0$, so no $\log\vee$ is left. The previous term $-2N_c\frac{1-\xi}{\xi}\log\frac{(y'-y)^2\mu^2}{4}\,d^3N_{\xi\,\mathrm{LO}}$ came from approximating $K(x,x',z)W(z,z)\to K(x,x',y)W(y,y)$ inside the z-integral. That approximation turns the finite integral into the IR-divergent $\int_z K(y,y',z)$, which produces the spurious $\log\mu$ and the asymmetry.

$$\frac{d^3N_{\mathrm{LO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\times\big[U^{ce}(y)-U^{ce}(x)\big]\;\rho^{e'}(x')\rho^{e}(x) \tag{3.14}$$

Definitions needed for consistency (Row IV). The Group II derivation contains no β_0, so the RC term is consistent only if $\mathcal{P}_{gg}$ in the DGLAP term is the full regularised splitting function

$$\mathcal{P}_{gg}(\xi)=2N_c\Big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\Big]+\frac{\beta_0}{2}\,\delta(1-\xi),\qquad \beta_0=\frac{11}{3}N_c\ \ \text{(pure glue)},$$

with the standard plus prescription on $[0,1]$, $\int_a^1 d\xi\frac{h(\xi)}{(1-\xi)_+}=\int_a^1 d\xi\frac{h(\xi)-h(1)}{1-\xi}+h(1)\log(1-a)$ for $a=k^+/\vee$. Then (i) the $\delta(1-\xi)$ term of DGLAP cancels the RC term exactly, reproducing the Group II result, which has no β_0. (ii) The $h(1)\log(1-k^+/\vee)$ piece is the $\log(1-k^+/\vee)$ part of the endpoint $\log\frac{(\vee-k^+)(k^++\Lambda)}{\vee\Lambda}$ (its $\log\frac{k^+}{\Lambda}$ part is in the JIMWLK term). Its pole, $+\frac{1}{(2\pi)^3}\frac{g^4N_c}{4\pi^4}\frac{1}{k^+}\frac{1}{\epsilon}\log(1-\frac{k^+}{\vee})\int\frac{(x-w)\cdot(x'-w')}{(x-w)^2(x'-w')^2}S$, cancels exactly the pole $\big[-\frac{2}{\epsilon}\big]\log(1-\frac{k^+}{\vee})$ of N_1 in Group I, Row II (with its c.c.), where $S=[U^{ab'}(x')-U^{ab'}(w')]\rho^{b'}(x')[U^{ab}(w)-U^{ab}(x)]\rho^{b}(x)$. So the only poles left are the DGLAP (collinear) pole and the RC pole. If instead $(1-\xi)_+$ were subtracted only on $[k^+/\vee,1]$, the term $-\frac{g^2}{8\pi^2}2N_c\log(1-\frac{k^+}{\vee})\big(\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}\big)\frac{d^3N_{\mathrm{LO}}}{d^3k}$ would have to be added explicitly.

4 Group III (with c.c.)

4.1 Row I

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left.-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times$$
$$\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y)-U^{d'b}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{4.1}$$

4.2 Row II

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right]$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times$$
$$\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{c'b}(z)\,\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{4.2}$$

4.3 Row III

4.3.1 First part

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_++\frac{1}{(p^++k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\left[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}$$
$$\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\left[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.$$
$$\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\left[U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{16\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{4.3}$$

4.3.2 Second part

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\\
&\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)^2}\,\left[(x-z)^2-(x-w)^2\right]+\frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.\\
&\left.-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]+\frac{\delta_{k'm}}{(p^++k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2}+\frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]\\
&-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_+\!+\frac{1}{(p^++k^+)}\right)\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2}+\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]\\
&\left.-\frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2}-\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2}+\frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times\\
&\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m}{(y'-z)^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}\\
&\left[-\frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times\\
&\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.\\
&\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\left[f^{c'd'a}\,U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(\right.\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-z)^{k'}}{(w'-z)^2\,(x'-z)^2}\\
&\left[\frac{(x-w)^2}{(x-z)^2\,(p^++k^+)\,[p^+\,(x-z)^2+k^+\,(x-w)^2]}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times\\
&\left[f^{c'd'a}\,U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.\\
&\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\\
&\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{cb}(z)\,U^{de}(x)\,\rho^e(x)\right]
\end{aligned}
\tag{4.4}
$$

4.4 Row IV

NA

4.5 Row V

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_++\frac{1}{(p^++k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\,\Big[U^{c'e'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]\\
&+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}\\
&\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\,\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\Big[U^{c'e'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right.\\
&\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\Big[U^{c'e'}(z)\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right]\times\\
&\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'ca}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{c'e'}(y')\,\rho^{e'}(x')-U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y)-U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\big]
\end{aligned}
\tag{4.5}
$$

4.6 Row VI

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left(\left[\frac{k^+}{p^+\,(p^++k^+)}\right]_++\frac{1}{(p^++k^+)}\right)\delta_{jm}\,\delta_{ik'}\right]\,\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]\\
&+\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}\\
&\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\,\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\,\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right.\\
&\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\,\Big[U^{c'e'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\right]\times\\
&\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]\\
&-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{c'e'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{d'b}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\big[U^{cb}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x)-U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\big]
\end{aligned}
\tag{4.6}
$$